

FORENSIC ANALYSIS REPORT

Image: jo-favorites-usb-2009-12-11.E01 • 2026-02-23 11:00:41 • Model: meta-llama/llama-3.1-8b-instruct

Risk Level: MEDIUM

EXECUTIVE SUMMARY

The analysis reveals evidence of timestomping and shadow copy deletion, indicating potential anti-forensic activity. Further investigation is required to determine the extent of the tampering and to verify the integrity of the file system.

Analysis completed in 10.0 seconds. AI analysis identified 3 findings. Anti-forensic analysis detected 260 potential indicators.

AI ANALYSIS FINDINGS

Finding #1: timestomping	MEDIUM	80% conf
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Evidence: Future timestamps found: 1, Sample future dates: ['USN journal missing']
Explanation: The presence of future timestamps, particularly the missing USN journal, is indicative of timestomping. This
Recommendation: Verify the integrity of the file system and investigate the USN journal for any signs of tampering.

Finding #2: shadow_deletion	HIGH	90% conf
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Evidence: \$Extend metadata exists: False, System Volume Information folder: False
Explanation: The absence of the \$Extend metadata and the System Volume Information folder is a strong indication of sh
Recommendation: Investigate the system for any signs of Volume Shadow Copy service tampering or deletion.

Finding #3: deletion	LOW	40% conf
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Evidence: Orphaned files: 1
Explanation: The presence of an orphaned file is not necessarily indicative of anti-forensic activity, as it could be a result
Recommendation: Verify the file system for any signs of file deletion or tampering.

ANTI-FORENSIC ANALYSIS

TIMESTOMPING DETECTION [256 indicators]

Detects manipulation of file timestamps

- Partition 0: Zero timestamp detected: r/r 3: JO FAVORITE (Volume Label Entry) 2009-11-20 09:30:34 (IST) 0000-00-00 00:
 - Partition 0: Zero timestamp detected: r/r 5: ._Trashes 2009-11-17 16:45:26 (IST) 2009-11-20 00:00:00 (IST) 0000-00-00
 - Partition 0: Zero timestamp detected: r/r 6: DSC00009.JPG 2009-11-11 09:16:56 (IST) 2009-11-24 00:00:00 (IST) 0000-00-
 - Partition 0: Zero timestamp detected: d/d 8: .Trashes 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000-00-00 0
 - Partition 0: Zero timestamp detected: + d/d * 133: ._02 2009-11-20 09:25:26 (IST) 2009-11-20 00:00:00 (IST) 0000-00-00
 - Partition 0: Zero timestamp detected: + r/r * 135: ._502 2009-11-20 09:25:28 (IST) 2009-11-20 00:00:00 (IST) 0000-00-0
 - Partition 0: Zero timestamp detected: + r/r * 136: ._502~1 2009-11-17 16:45:28 (IST) 2009-11-17 00:00:00 (IST) 0000-00-
 - Partition 0: Zero timestamp detected: + r/r * 138: ._501 2009-11-17 16:45:28 (IST) 2009-11-17 00:00:00 (IST) 0000-00-0
 - Partition 0: Zero timestamp detected: d/d * 10: .fsevents 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000-00
 - Partition 0: Zero timestamp detected: d/d 13: .Spotlight-V100 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000
 - Partition 0: Zero timestamp detected: + d/d 646: Store-V1 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000-00-
 - Partition 0: Zero timestamp detected: ++ r/r 775: VolumeConfig.plist 2009-11-17 16:45:28 (IST) 2009-11-20 00:00:00 (IST) (IS
 - Partition 0: Zero timestamp detected: ++ d/d 777: Stores 2009-11-17 16:45:28 (IST) 2009-11-17 00:00:00 (IST) 0000-00-0
 - Partition 0: Zero timestamp detected: +++ d/d 1544: BC650BA8-C5B5-4A64-8BDB-18A9533BF1D7 2009-11-17 16:45:28 (IST) 200
 - Partition 0: Zero timestamp detected: ++++ r/r 1669: psid.db 2009-11-20 09:30:34 (IST) 2009-11-20 00:00:00 (IST) 0000-
- ... and 241 more

SHADOW COPY DELETION [Clear]

Evidence of VSS deletion attempts

No indicators found in this category.

HIDDEN DATA STREAMS (ADS) [Clear]

Alternate Data Streams detection

No indicators found in this category.

LOG CLEARING EVIDENCE [2 indicators]

Evidence of log file clearing

- Partition 0: No Windows event log files (.evtx) found in listing
- No Windows Event Logs (.evtx) found in extracted logs directory

REGISTRY TAMPERING [Clear]

Registry modification indicators

No indicators found in this category.

FILE DELETION TRACKING [1 indicator]

Deleted file recovery opportunities

- Partition 0: Orphaned files found (deleted but recoverable): 1 entries

MFT ANOMALIES [1 indicator]

Master File Table irregularities

- Partition 0: MFT contains 4 system file entries

SUMMARY STATISTICS

Category	Indicators
Timestamping	256
Shadow Copy Deletion	0
Hidden Streams (ADS)	0
Log Clearing	2
Registry Tampering	0
File Deletion	1
MFT Anomalies	1
TOTAL	260

Overall Anti-Forensic Risk Level: HIGH

RECOMMENDATIONS

1. Verify the integrity of the file system
2. Investigate the USN journal for signs of tampering
3. Investigate the system for any signs of Volume Shadow Copy service tampering or deletion

Additional Anti-Forensic Investigation Steps:

- Conduct detailed timeline analysis around detected anomalies.
- Examine Windows Event Logs for evidence of vssadmin or wevtutil commands.

- Analyze \$MFT and \$UsnJrnl for file system manipulation evidence.
- Review Registry hives for persistence mechanisms and tool artifacts.
- Search for known anti-forensic tool signatures and artifacts.